

Stochastic Calculus for Quants

Intuition, Memory Tricks, and Desk-Level Logic
From Brownian Motion to Pricing

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Module 4: SDEs You Actually See on Desks—Modeling Choices

4.1 The SDE Menu: What Models What

SDE Selection Decision Tree "Choosing drift/diffusion = choosing economics" $dX_t = \mu(X_t, t)dt + \sigma(X_t, t)dW_t$ "Interview: 'I choose GBM for equity because it ensures positivity. I choose OU for rates because they mean-revert. I choose CIR to enforce positivity in vol.'"

Core Formula

Model	SDE	Used for	Key Intuition
GBM	$dS = \mu S dt + \sigma S dW$	Equity/FX	Proportional randomness
OU	$dX = \kappa(\theta - X)dt + \sigma dW$	Rates, spreads	Rubber band mean-reversion
CIR	$dX = \kappa(\theta - X)dt + \sigma\sqrt{X}dW$	Rates, intensity	Positivity + mean-reversion
Heston	$dv = \kappa(\theta - v)dt + \xi\sqrt{v}dW$	Stochastic vol	Vol clusters + leverage

4.2 Geometric Brownian Motion (GBM): Equity Workhorse

SDE:

Core Formula

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

Properties:

- **Positivity:** $S_t > 0$ almost surely (lognormal)
- **Scale-invariance:** Returns independent of level
- **Closed-form:** $S_T = S_0 \exp\left((\mu - \frac{1}{2}\sigma^2)T + \sigma W_T\right)$

"GBM = Compounding + Noise" **Memory:** Drift μS_t = exponential growth. Diffusion σS_t = percentage volatility constant. **Fails for:** Rates (can go negative), vol (clusters), credit spreads (mean-revert).

When GBM Fails: - Rates near zero → need positive models (CIR) - Vol spikes → need stochastic vol (Heston) - Credit spreads → need mean-reversion (OU)

GBM Strengths "Simple, tractable, ensures positivity" $\log S_T \sim N(\log S_0 + (\mu - \frac{1}{2}\sigma^2)T, \sigma^2 T)$ "Interview: 'GBM is the default for equity because it matches what we see: log returns are roughly normal, and prices stay positive.'"

4.3 Ornstein-Uhlenbeck (OU): Mean-Reverting Spreads

SDE:

Core Formula

$$dX_t = \kappa(\theta - X_t)dt + \sigma dW_t$$

Properties:

- **Mean-reversion:** κ = speed of reversion to θ
- **Stationary:** Long-run distribution $N(\theta, \sigma^2/2\kappa)$
- **Can go negative:** Not for rates

4.6 Model Selection Decision Tree: "Which SDE Do I Start With?"

"Start With Economics, Not Math" **Decision Flow:**

1. **Must prices stay positive?**
 - Yes → GBM (equity/FX) or CIR (rates/intensity)
 - No → OU (spreads, temperature)
2. **Mean-reversion needed?**
 - Yes → OU or CIR (rates, spreads, vol)
 - No → GBM (equity, commodities)
3. **Volatility constant?**
 - Yes → GBM or OU (simple)
 - No → Heston (stochastic vol), SABR (vol smile)
4. **Speed vs realism?**
 - Fast → GBM (closed-form)
 - Accurate → Heston (calibrate to surface)

Common Mistake: Using GBM for rates because it's "simple". Result: negative rates in your model while market never trades negative.

SDE Selection Heuristic "Economics → Math, not math → economics" Choose model that captures required features (positivity, mean reversion, vol clustering) Interview: 'I start with the economic features I need: positivity? mean reversion? vol dynamics? Then I pick the simplest SDE that captures them.'

4.7 Parameter Interpretation: What Do They Mean?

Parameter	Model	Economic Meaning	What Happens If Large
κ (speed)	OU, CIR, Heston	How fast mean reversion is	Large = fast snap back, small = persistent
θ (long-run)	OU, CIR, Heston	Equilibrium level	Shifts center of distribution
σ (vol of noise)	GBM, OU, CIR	Base volatility	Larger = fatter tails, more uncertainty
ξ (vol of vol)	Heston	How volatile volatility is	Large = more smile, fatter tails
ρ (correlation)	Heston	Leverage effect	Negative = crash risk priced in

"Big = Panic, Big = Speed Bump" **Memory:** High σ means the process is jumpy. High κ means it quickly returns to normal (e.g., VIX after spike).

Parameter Sensitivity "Parameters have direct economic meaning" κ = patience, θ = gravity, σ = chaos, ξ = panic, ρ = fear Interview: 'If I calibrate Heston and get $\kappa=5$, that's a very fast mean-reverting vol. If $\rho=1.0$, that's high vol-of-vol → expect fat tails.'

tion costs: continuous hedging infinite trades. (2) Discrete time: we hedge daily/hourly, not continuously. (3) Gap risk: prices can jump (e.g., overnight) between hedges. The hedging error is proportional to $\Gamma^2 \sigma^4 \Delta t$ and transaction costs dominate at high frequency."

12. What is the market price of risk? Girsanov drift. Answer: "It's the extra return per unit of volatility that investors demand to hold risk. In measure change, it shows up as $\theta = (\mu - r)/\sigma$. When moving from physical to risk-neutral measure, we subtract θdt from the Brownian motion, which changes the drift of all assets to r . It's not directly observable but is implied by option prices."

13. How does Itô's Lemma help with Greeks? Derivation. Answer: "Applying Itô to $V(S_t, t)$ directly gives the PnL decomposition: $dV = \Theta dt + \Delta dS + 0.5\Gamma(dS)^2$. The coefficients *are* the Greeks. Itô's Lemma is the mathematical justification for Taylor expansion of option prices under randomness. It also shows why Theta and Gamma are linked via the PDE."

14. When would you use OU vs CIR? Positivity constraint. Answer: "Use OU when the variable can be negative (e.g., spreads, temperature). Use CIR when positivity is required (e.g., interest rates, default intensities, variance). CIR's $\sqrt{X}$ diffusion prevents zero crossing if Feller condition holds. OU is simpler but can produce negative values which are economically nonsensical for rates."

15. Explain risk-neutral pricing without formulas. Smaller $\Delta t \rightarrow$ better but more cost.

"Theory: Continuous. Reality: Daily + Reserve" Memory: Market makers hedge intraday for gamma, daily for delta. They hold reserve for gap risk. Cost of hedging = transaction costs + capital cost of reserve.

Friction Management"Hedge what matters, reserve for what you can't"Reserve = 10-20% of gamma exposure $\times$ jump size"Interview: 'We accept discrete hedging error and hold reserves for jumps. Perfect hedging is too expensive.'"